

TrinPM20

(0:18 - 0:37)

This training instruction shows how to calibrate the INGICOM Sound Power Max inverters. The first thing you have to do is to access the inverter following the indications given in a previous training instruction. This time, however, we will close the auxiliary thermal magnetic breaker.

(0:38 - 0:58)

This will power up the display and electronic blocks. The emergency push button should be activated at the beginning of the calibration. In this particular example, the electronic block we are going to calibrate is located in the second DC-AC conversion module.

(0:58 - 1:15)

The next thing we have to do is to open the first cabinet where the AC-DC switches are located. We will proceed to close the second DC breaker that is dedicated to that particular module. We will use a wrench to do so.

(1:18 - 1:41)

After that, we will close the AC thermal magnetic breaker so that module gets both AC and DC power. Sometimes, this thermal magnetic breaker is not found within the inverter and it can be found elsewhere in the shelter. We recommend you check that there is a cable that goes from the neutral in the inverter to the neutral in the transformer.

(1:44 - 2:11)

If we check the display we can get to see that the inverter is getting both AC and DC power and if we select the second module we can actually see that this particular one is getting powered. We will start calibrating the DC voltage in the second electronic block. We will use a class 3 multimeter to measure the DC voltage between the two terminals that you can see on the image.

(2:16 - 2:35)

The reading we get is around 506 volts DC. This reading must be the same as the one we get in the display. As you can see in the image, this reading in the display is slightly different to the one we got in the multimeter.

(2:36 - 2:48)

507 volts DC in this case. That means that we have to calibrate the second module by changing

or modifying the DC scale factors. We access the main menu and then we go to settings.

(2:50 - 3:32)

Then we go to config adjustments and we finish typing the password that we have been given. 3725 When we access the next menu we will need to select parameters change. The first values we can see represent the AC voltage, the next one represent the AC frequency and if we keep going down by pressing the OK button we can get to the point where we can modify the DC voltage scale factors.

(3:33 - 4:02)

In this particular example we are going to make a slight modification of the DC voltage for the negative input by pressing the arrows. In order to exit from this menu we have to keep pressing OK till we get to save the new configuration. If we go to the main screen we can get to see how the DC voltage has changed and what we have to do is to verify once again with the actual reading we get from the multimeter.

(4:14 - 4:32)

Now it is time to calibrate the AC voltage. The terminals we have to check are the ones shown in the image. The first one is J6 which is for line number 1, the next one is J7 for line number 2 and the third one J8 has two cables connected.

(4:32 - 5:01)

The one on top is for neutral and the one on the bottom is for line number 3. The measurements with the multimeter will always be made between neutral and each of the phases. In this case phase number 1 is presenting a AC voltage equal to 161.6 volts AC. As you can see in the image the display is showing a slight different value.

(5:02 - 5:20)

Therefore we will need to calibrate the AC readings for that particular electronic block. We need to access the main menu, then settings and after that we access other adjustments. We will be asked to enter the same password as before, 3725.

(5:22 - 5:46)

We access scale factor adjustments in node number 2. The first values we get are for the AC phases and we will start modifying the first of the phases. We press modify and we will enter the next screen. Before we enter the actual values we recommend you use the multimeter to get the most accurate readings possible.

(5:50 - 7:07)

Once we know the value of these readings for each of the phases we proceed to input them through the display. We will use the arrows to increase or decrease the values that are shown on the display. After pressing ok we strongly recommend that you make a new verification by reading once again the AC voltages using the multimeter and verifying that the same values are shown on the display.

(7:26 - 7:44)

Now that we know that this electronic block is reading the AC voltage properly it is time to calibrate the DC current measurements. We access the main menu, then settings and other adjustments. We will be asked to enter the previously specified password.

(7:45 - 8:00)

Once we are in the next screen we have to access mode inverter menu. When we adjust the DC current we need a constant energy flow towards the inverter. That's why this process is only recommended on sunny days with no clouds.

(8:00 - 8:29)

We need an irradiation of at least 700 watts per square meter or we need that the inverters around the one we are working with can produce at least 70% of their nominal power. We will select consign mode menu and then we will press yes. The so-called consign operating mode forces the electronic block to work at an specified level.

(8:30 - 8:41)

We are going to start selecting 10%. Then we release the emergency push button. Your inverter will probably have a sensor on top of the door that indicates when the door is open.

(8:42 - 9:05)

In that case you will need to bypass it in order to carry on with the calibration. That's not the case of the inverter that we are using for this video. Now it is time to change the power that the inverter is producing to 60% and we will do that by pressing the arrow pointing up and the OK button at the same time.

(9:06 - 9:23)

We can clearly see in the display that the power that is producing our inverter is rising. When it gets stable it is time to take the wattmeter and place it in the right location. We have selected the option to measure DC currents.

(9:27 - 9:47)

The wattmeter is now placed in the negative DC input. This value should correspond with the

one we get in the display. To match these two values we go to the main menu then to settings and other adjustments.

(9:57 - 10:19)

When we enter the password we will be given access to a new screen where we have to select scale factor adjustments. The parameter that we might need to modify is the one called IPV. In this case the value we are getting is 134 whereas the real one should be 141.

(10:21 - 10:42)

Therefore we select modify and we enter a new value. The process will be finished when we find a new value that actually makes the multimeter reading and the display reading match. The process to find this new value takes some time and it is a matter of test and error.

(10:44 - 11:22)

Little by little we will manage to get the display to read the actual values. Remember that if you press the arrows and the OK button at the same time you will be able to change the values shown on the display in a faster way. When the values we get in the display are close to the ones we get in the multimeter we can consider that the process is finished.

(11:23 - 11:43)

Now it is time to calibrate the next variable that the inverter is measuring, the AC currents. Take care when you manipulate the wattmeter because we are going to place it on the contactor which is being fed and there is power in it. We need to compare the reading we get in the wattmeter with the one the display is showing.

(11:43 - 12:03)

If there is a difference between them what we recommend is that you change the scale factor of that particular phase through the display. In this case the IAC1 value that the display is showing is far larger than the one in the multimeter. Therefore we will need to modify it by pressing the OK button and the arrows.

(12:11 - 13:37)

When we finish calibrating that phase we proceed to do the same thing with the next phase, phase number 2. The same needs to be done with the last of the phases, phase number 3. Remember that if you press the arrow pointing down and the OK button at the same time you decrease the value of the variable much faster. Same as before we recommend that you check that the calibration is well made. To do so you have to measure the AC current in each of the phases through the multimeter and you verify these values with the ones in the display.

(14:00 - 14:27)

Before we continue we want to remind that in the main menu you can select monitoring and there you can get to see some of the most important variables that affect the working conditions of the inverter. The calibration of the AC current has finished and now it is time to calibrate the reactive power. The calibration of the reactive power is a process that needs to be made in two different production points.

(14:27 - 14:42)

The first production point we are going to select the display is 20%. To do so we go to the main menu, settings, other adjustments and then we enter the password. When we are in we have to select mode inverter menu.

(14:43 - 15:00)

There we will select consign mode menu. When you enter that menu you can get to see that the inverter was being forced to work at 60%. We are going to decrease that value through the display and we are going to force to work the inverter at a 20% of its nominal power.

(15:04 - 15:21)

The power that the inverter is injecting drops down slowly until it gets to a stable point. Only when we reach that point we can calibrate the reactive power. We can now exit all these menus by pressing escape.

(15:24 - 15:53)

We take our wattmeter and we select three phase power readings. When the wattmeter is ready to measure three phase powers it gives you a recommendation on the main screen that we will follow. The clamp will be placed around the first phase to measure the current and the other two elements will be placed in the other two phases to read the voltage.

(15:54 - 16:28)

Remember that these terminals will be protected by a plastic and you will need to remove it before you conduct this calibration. The wattmeter is telling us that the active power is around 31.8 kW and we also have some negative reactive power. Due to the fact that the inverter has been set not to inject any reactive power we have to get that reactive power that we see in the wattmeter equal to zero.

(16:28 - 16:52)

Normally that zero value is difficult to get so we want it to be fluctuating between positive and negative values. To do this we will go to the main menu in the display and we will select settings, other adjustments and then we will enter the password. Then we select scale factor adjustments and we will get to see the node we are working with and then we press ok.

(16:54 - 17:13)

What we need to do is to move to the following screens by pressing the arrow pointing down. When we get to the screen where we see the reactive power X1 we stop there and we modify it. The first value you see blinking on the screen is the one we have to modify.

(17:15 - 18:04)

It is difficult to know how much you have to modify so what we recommend is that you keep trying till you get to a value where the reactive power gets in a point that is between positive and negative values. As you can see from the image this is a process that takes some time. However we have to say that with experience you will be able to do these things much faster.

(18:07 - 18:37)

As you can see from the image now the reactive power is fluctuating between positive and negative values. When you get to that point you can conclude that the calibration of the reactive power at 20% of the nominal power has been done. However we have to repeat the same process but this time at 70% of the nominal power.

(18:37 - 18:56)

Therefore we will need to increase the power that the inverter is producing. To do so we will go to the display and in the main menu we will select settings, other adjustments and then we will enter the password. We go once again to mode inverter menu and then to consign mode menu.

(18:56 - 19:17)

When we get there we can now increase the power that the inverter is producing to 70%. When the power produced by the inverter is stable we can now exit once again to the main menu. There we will select settings, other adjustments and then we will enter the password again.

(19:29 - 20:02)

This time instead of going to reactive power X1 we have to go to X2 Y2 because we want to define the second point in the graph for the calibration of the reactive power. If we check the wattmeter we can now see that the reactive power that the inverter is injecting is negative and it's around 7 or 8 kVA. We have to calibrate the inverter to avoid this situation because we want the reactive power to be fluctuating around 0 kVA.

(20:02 - 21:56)

Once again this process might take some time and it is a question of trying and trying again until we get the reactive power injected in the grid equal or near 0. As you can see in the image the reactive power now is fluctuating between positive and negative values so we can conclude

that the reactive power of the inverter has been calibrated. The final adjustment we have to make to the inverter is the one we call power consumption. To do so we have to go back to Consume Mode menu and there we have to select 0 so that means that the inverter won't be injecting any power on the grid.

(21:56 - 22:35)

At this point the inverter shouldn't be injecting any power on the grid and actually if you check the wattmeter or the display there is a negative value that represents the power that the inverter is consuming. This value needs to be calibrated as well. What we have to do is to go to the main menu, Settings, Other Adjustments and then Scale Factor Adjustments.

(22:44 - 23:28)

The parameter that is blinking on the screen, on the display, needs to be matched to the one we can see at the wattmeter. The consumption power has now been adjusted and what we have to do is to verify that this adjustment has been done properly. To do so what we recommend is that you increase the power that the inverter is producing and then you check that the active power that is shown on the display corresponds to the one that is shown on the wattmeter.

(23:29 - 23:52)

In this example we are going to make the verification in three different points. 20% of the nominal power, 50% of the nominal power and 70% of the nominal power. The maximum error we can accept in the display is 3% of the value that the wattmeter is measuring.

(24:45 - 24:59)

As you can see in the video the 3% error that we accept has never been reached in any of the points where we have made the verification. With this final compromise we conclude the calibration process of the inverter.